

CHAPTER – 1. UNITS AND MEASUREMENT

Important formulas and concepts directly from NCERT-

➤ Angle,

$$\theta = \text{arc} / \text{radius}$$

❖ Systematic error-

- Errors which tend to be in one direction , either positive or negative .

❖ Random errors-

- Errors which occur regularly and hence are random with respect to sign and size.

❖ Least count-

- The smallest value that can be measured by measuring by the measuring instrument is called its least count.

- It is the error associated with the resolution of the instrument.

❖ Significant figures-

- All non-zeroes are significant.

- Zeroes b/w non-zeroes are significant.

- Zeroes at the end of non-decimal figures are non-significant.
(e.g. 20300 , S.F. = 3)

- Zeroes at the end of decimal figures are significant.
(e.g. 0.409800 , S.F. = 6)

- Zeroes present at starting of decimal figures are non-significant.
(e.g. 0.0057800, S.F. = 5)

❖ Rules for Arithmetic Operations with Significant Figures(S.F.)-

(1) In multiplication or division, final result should retain as many as S.F. as in original no. With least S.F.

(2) In addition or subtraction, final result should retain as many as decimal places as are in no. with least decimal places.

SUMMARY

1. Physics is a quantitative science, based on measurement of physical quantities. Certain physical quantities have been chosen as fundamental or base quantities (such as length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity).
2. Each base quantity is defined in terms of a certain basic, arbitrarily chosen but properly standardised reference standard called unit (such as metre, kilogram, second, ampere, kelvin, mole and candela). The units for the fundamental or base quantities are called fundamental or base units.
3. Other physical quantities, derived from the base quantities, can be expressed as a combination of the base units and are called derived units. A complete set of units, both fundamental and derived, is called a system of units.
4. The International System of Units (SI) based on seven base units is at present internationally accepted unit system and is widely used throughout the world.
5. The SI units are used in all physical measurements, for both the base quantities and the derived quantities obtained from them. Certain derived units are expressed by means of SI units with special names (such as joule, newton, watt, etc).
6. The SI units have well defined and internationally accepted unit symbols (such as m for metre, kg for kilogram, s for second, A for ampere, N for newton etc.).
7. Physical measurements are usually expressed for small and large quantities in scientific notation, with powers of 10. Scientific notation and the prefixes are used to simplify measurement notation and numerical computation, giving indication to the precision of the numbers.
8. Certain general rules and guidelines must be followed for using notations for physical quantities and standard symbols for SI units, some other units and SI prefixes for expressing properly the physical quantities and measurements.
9. In computing any physical quantity, the units for derived quantities involved in the relationship(s) are treated as though they were algebraic quantities till the desired units are obtained.
10. In measured and computed quantities proper significant figures only should be retained. Rules for determining the number of significant figures, carrying out arithmetic operations with them, and 'rounding off' the uncertain digits must be followed.
11. The dimensions of base quantities and combination of these dimensions describe the nature of physical quantities. Dimensional analysis can be used to check the dimensional consistency of equations, deducing relations among the physical quantities, etc. A dimensionally consistent equation need not be actually an exact (correct) equation, but a dimensionally wrong or inconsistent equation must be wrong.